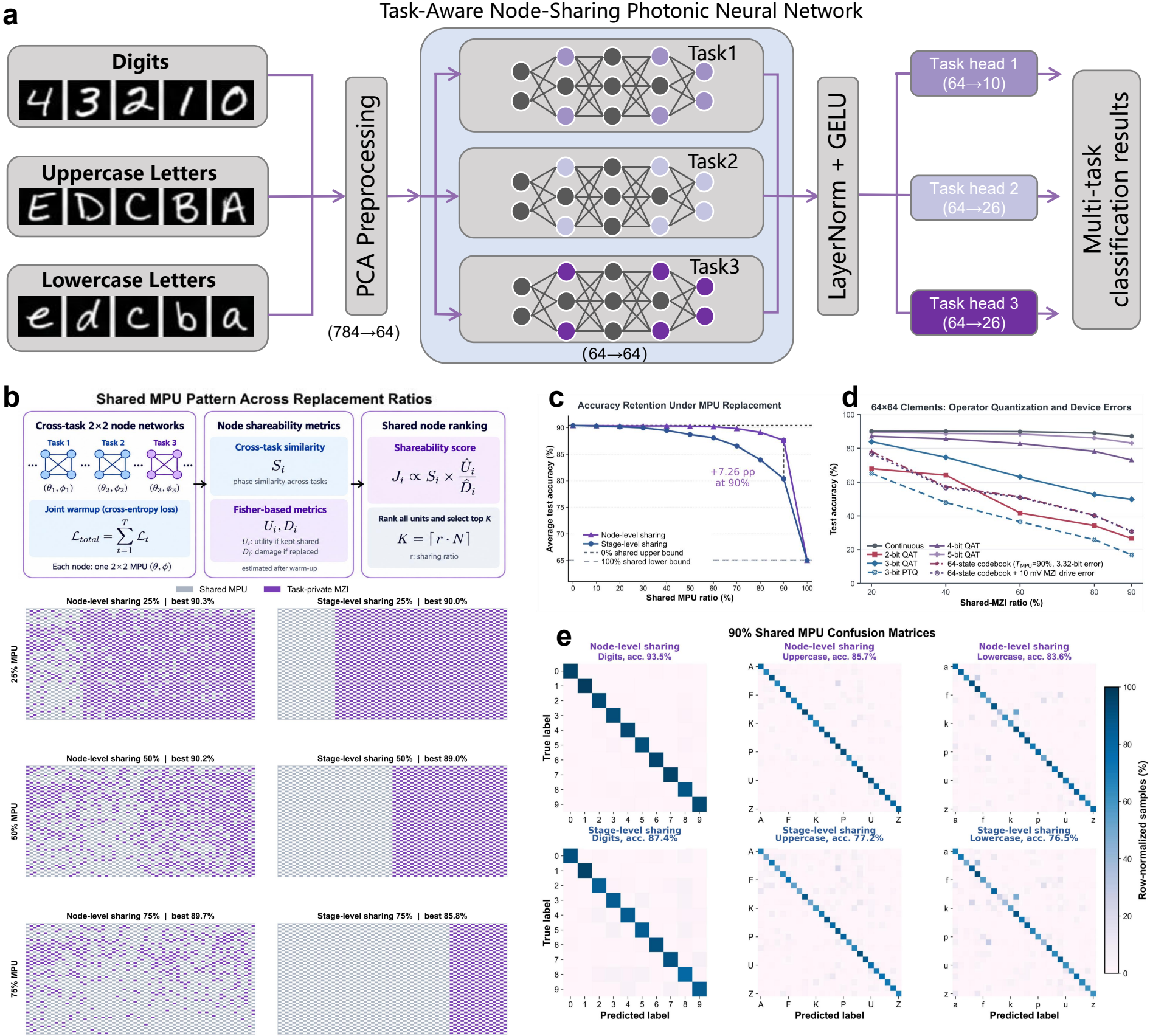

a
Task-Aware Node-Sharing Photonic Neural Network
Digits
Uppercase Letters
Lowercase Letters
PCA Preprocessing
(784→64)
Task1
Task2
Task3
(64→64)
LayerNorm + GELU
Task head 1 (64→10)
Task head 2 (64→26)
Task head 3 (64→26)
Multi-task classification results
b
Shared MPU Pattern Across Replacement Ratios
Cross-task 2×2 node networks
Task 1
Task 2
Task 3
Joint warmup (cross-entropy loss)
Each node: one 2×2 MPU
Node shareability metrics
Cross-task similarity
phase similarity across tasks
Fisher-based metrics
estimated after warm-up
Shared node ranking
Shareability score
Rank all units and select top K
r: sharing ratio
Shared MPU
Task-private MZI
Node-level sharing 25% | best 90.3%
Stage-level sharing 25% | best 90.0%
Node-level sharing 50% | best 90.2%
Stage-level sharing 50% | best 89.0%
Node-level sharing 75% | best 89.7%
Stage-level sharing 75% | best 85.8%
25% MPU
50% MPU
75% MPU
c
Accuracy Retention Under MPU Replacement
Average test accuracy (%)
Shared MPU ratio (%)
+7.26 pp at 90%
Node-level sharing
Stage-level sharing
0% shared upper bound
100% shared lower bound
d
64×64 Clements: Operator Quantization and Device Errors
Test accuracy (%)
Shared-MZI ratio (%)
Continuous
2-bit QAT
3-bit QAT
3-bit PTQ
4-bit QAT
5-bit QAT
64-state codebook (T_MPU=90%, 3.32-bit error)
64-state codebook + 10 mV MZI drive error
e
90% Shared MPU Confusion Matrices
Node-level sharing
Digits, acc. 93.5%
Uppercase, acc. 85.7%
Lowercase, acc. 83.6%
Stage-level sharing
Digits, acc. 87.4%
Uppercase, acc. 77.2%
Lowercase, acc. 76.5%
True label
Predicted label
Row-normalized samples (%)